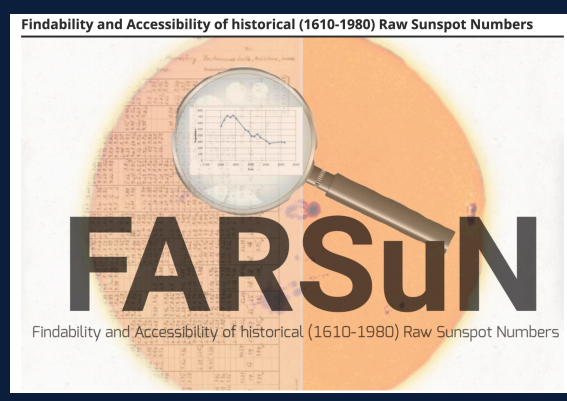




Reconstruction of the Sunspot Number Series : Gathering Data

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Abstract SH23E-2613* – 16 Dec 2025, 14:15–17:45

The Sunspot Number (SN) is one of the longest continuous records of solar activity. SN V2.0 significantly improved the historical series through targeted recalibrations, but recent studies have shown that some scale changes—such as the 1849 transition—can only be fully understood by returning to the original raw observations.

This motivates a complete reconstruction (SN V3.0) in which uncertainties and calibration steps arise directly from the quality of each dataset rather than from global statistical corrections.

Four-Axis Methodology

- Gather Sources:** Track down dispersed observing logs (Wolf's *Mittheilungen*, Zürich tables, Gruithuisen manuscripts, Adams drawings) and capture provenance metadata.
- Process Data:** Use handwriting/OCR engines (Transkribus models) alongside manual double-keying for fragile entries.
- Validate & Standardize:** Reconcile aliases, observer IDs, and calendars; model group/spot counting behaviour to derive calibration curves.
- Disseminate:** Publish machine-readable products, APIs, and VO-ready services with versioned uncertainty bundles.

FARSuN Historical Database

A sustainable, interoperable reference database ensuring that four centuries of solar observations remain accessible.

Present Status

- Legacy sources integrated for cross-validation and QC.
- Automated quality checks detect inconsistencies (e.g., Groups > Spots).
- Metadata harmonization for FAIR/VO standards aligned with EPN-TAP.

Next Steps

- Finalize canonical schema and close remaining legacy mappings.
- Scale up digitization (Adams, Gruithuisen, Stark) using HTR.
- Expose full dataset through EPN-TAP service.
- Foundation for SN V3.0 calibration.**

Technical Methods

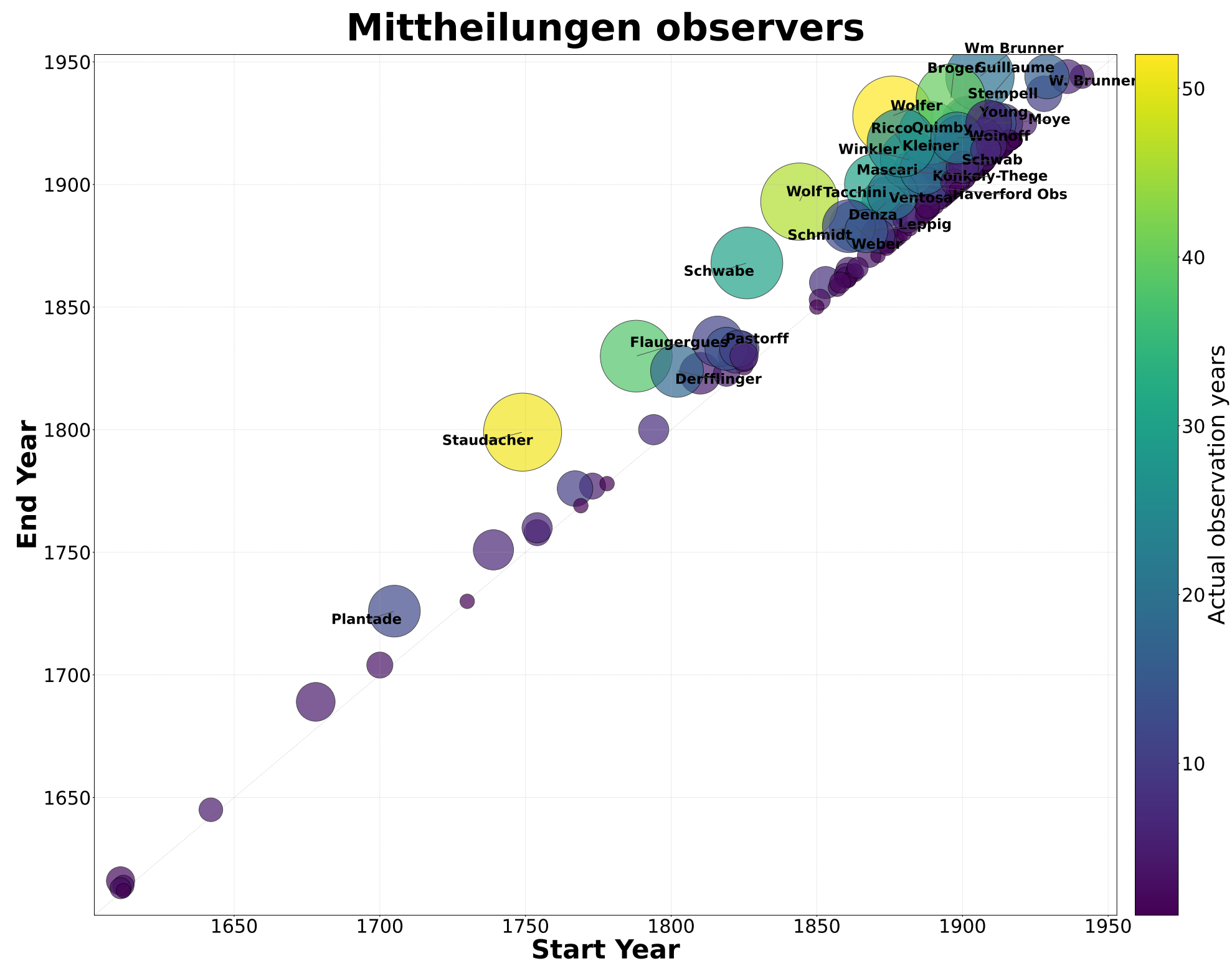
- Dual-mode transcription:** Tailored Transkribus handwriting models plus manual keying.
- Data hygiene:** Observer alias reconciliation, calendar conversions.
- Uncertainty modelling:** Propagate observer calibration curves and spotless-day confidence flags.
- Automation:** Reproducible Python pipelines with checkpoints.

Quality Gates

- Two-person verification** for Adams and Gruithuisen counts.
- Cross-compare** transcriptions with Wolf's *Mittheilungen* and modern indices.
- Git-tracked pipelines** capture every transformation for audit.

Key Datasets: The Core Collection

Mittheilungen Dataset (1610–1945)



The *Mittheilungen* journals, compiled by Rudolf Wolf and successors, form the central source of historical sunspot observations (1848–1945) and incorporate earlier material back to 1610.

Status: Fully digitized at ROB.

- Integrated into FARSuN DB.
- Refining consistency and observer calibration (Wolf–Wolfer overlap).

Zürich Tables (1945–1979)

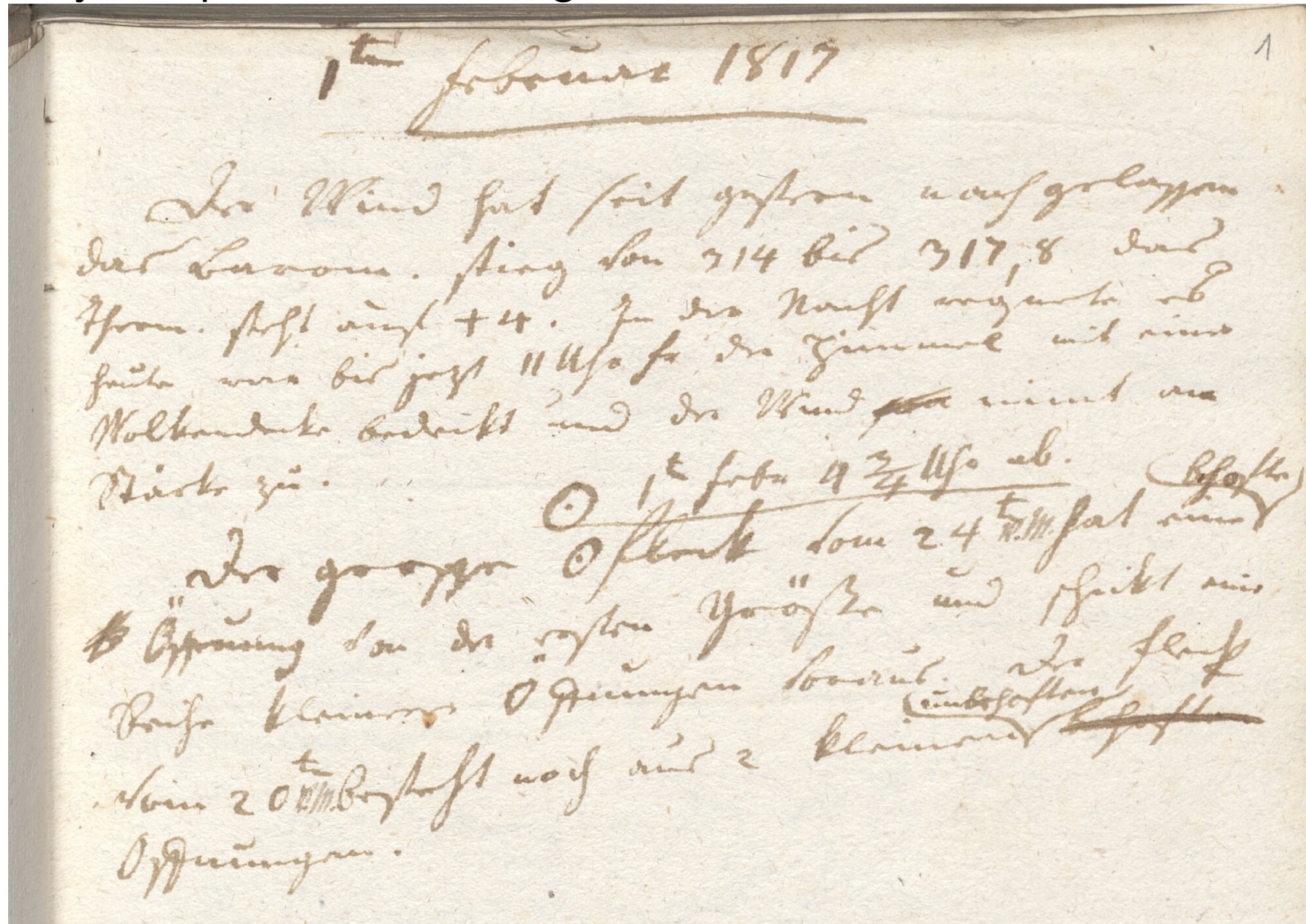
The foundational record of daily sunspot number production at Zürich.

Status: 95% of 2,000+ yearly tables digitized.

- Fully digitized for 1945–1979 with detailed metadata completed for 1945–1959.
- Quality control and observer metadata harmonisation underway.

Gruithuisen Data (1817–1849)

Daily sunspot notes, drawings, and weather context in German Kurrent script. Vital early-19th-century coverage.



Status: Images fully collected.

- Extraction via bespoke Transkribus models.
- Two-pass QC (survey + counting) in progress.

C. H. Adams Data (1819–1823)

FARSuN has recovered 338 full-disk drawings and 1,056 daily entries.

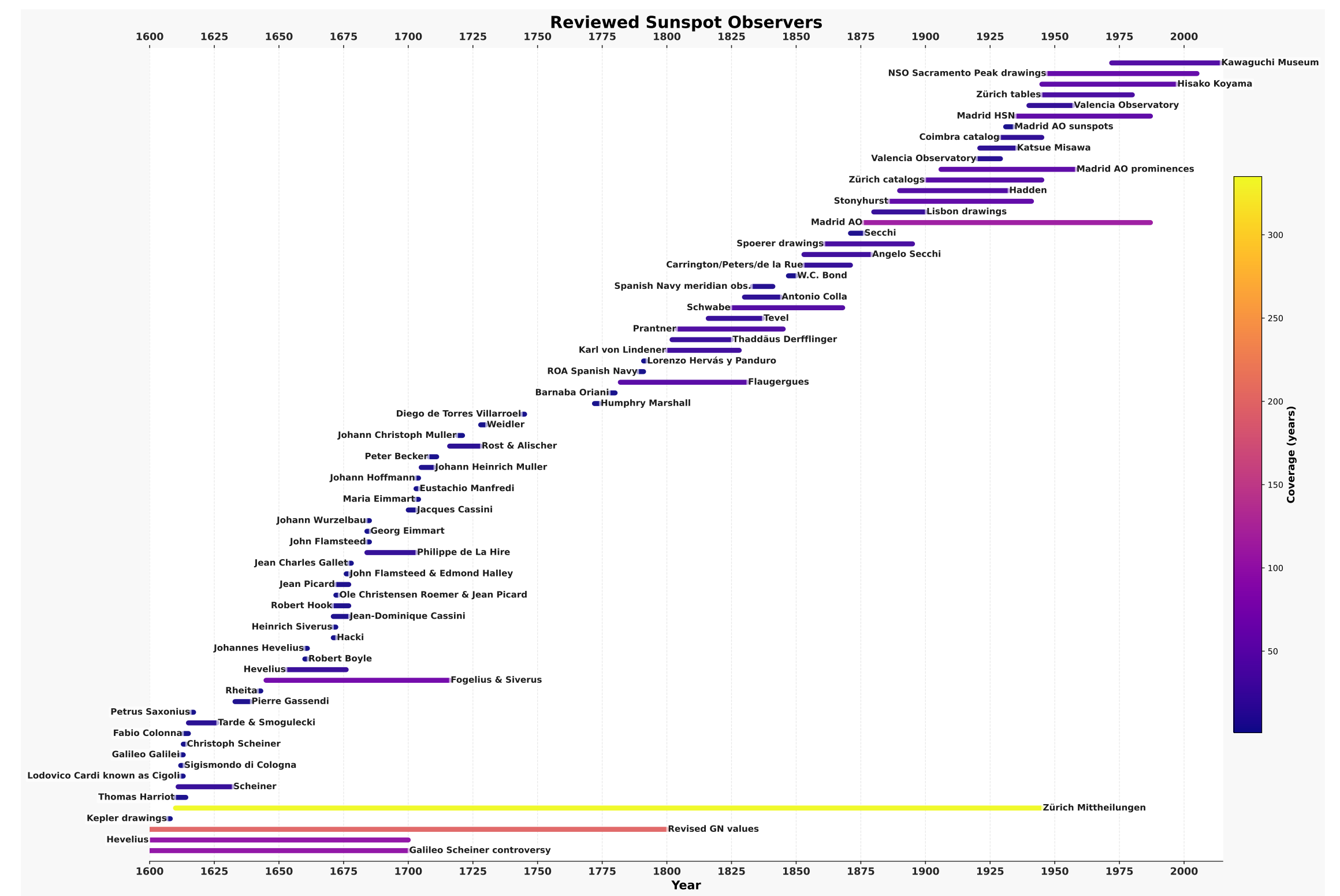
- Dense coverage with explicit spotless days.
- Cross-check:** Spotless tallies matched against Wolf's compilations to anchor early-1800s calibration.

Additional Data Recovery

Augustin Stark (1813–1835)

Printed German descriptions of sunspot groups. Preliminary OCR shows good potential; refining QC flags and metadata integration.

Timeline of Data Recovery (Since 2010)



Community efforts (Hayakawa, Arlt, Vaquero, Carrasco) have expanded available raw observations, enabling independent cross-checks.

From Raw to SN V3.0

- Curate daily counts with provenance-rich metadata.
- Solve observer response functions to align scales across centuries.
- Aggregate to indices with propagated uncertainties.
- Benchmark against SN V2.0 and satellite-era proxies.

Project Challenges

- Deciphering *Kurrent* manuscripts without losing context.
- Harmonising heterogeneous formats (drawings, ledgers).
- Quantifying observer bias and spotless-day reporting habits.
- Maintaining end-to-end provenance.

Space Climate Impact

Citizen Science

Portal mobilises volunteers to transcribe *Kurrent* & sustain STEM outreach.

Forecasting

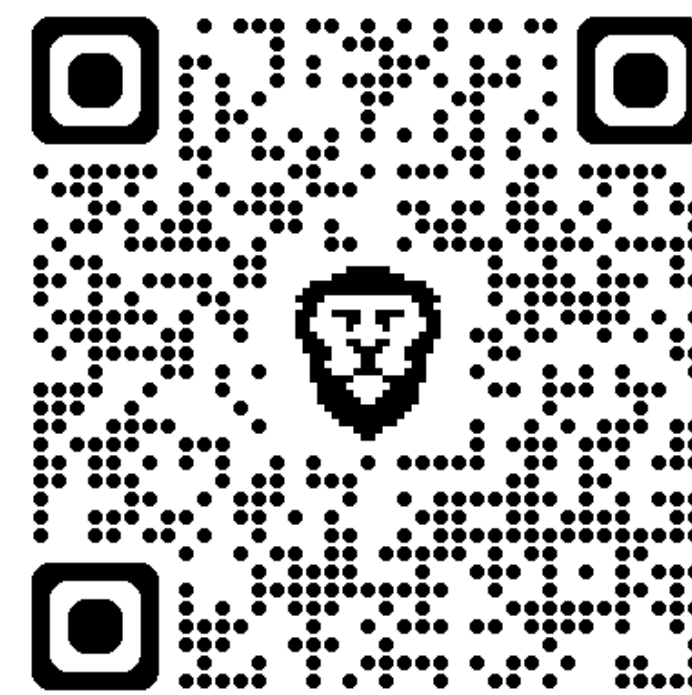
Bias-free SN V3.0 improves solar cycle prediction and operational baselines.

Climate Links

Validated historical forcing refines long-term climate models.

Open Science

FAIR-compliant releases with APIs and uncertainty bundles.



Scan for Data

For more information on the FARSuN project, datasets, and a digital copy of this poster.

Contact: sidc-support@oma.be

Portal: <https://www.sidc.be/silso/>